

**DERIVATION OF DESIGN DATA FOR
FIBROUS COMPOSITES**

Part: **1.4 – COMPOSITE MATERIALS**

Chapter: **1481**

Title: **DERIVATION OF DESIGN DATA FOR
FIBROUS COMPOSITES**

Revision: **1.2**

Date: **08/06/2023**

1 INTRODUCTION**1.1 OBJECT AND SCOPE**

This chapter completes the methodology discussed in chapter 1480 “DERIVATION OF DESIGN DATA IN COMPOSITES” for generating the allowables in case the material is a fibre reinforced fabric.

2 PHYSICAL PROPERTIES**Ply thickness**

Nominal ply thickness shall be a unique scalar value defined by M&P, to be used by the different functions of the Design Office in models and analyses, and be representative of real production parts (it shall be also the reference for quality control of real production parts).

Coefficients of thermal expansion

The coefficients of thermal expansion shall be presented in terms of mean values, for material directions 1 (longitudinal), 2 (transversal) and 3 (through-thickness, if available), as function of the temperature. A typical format of presenting these data shall be averaged values valid for application in within typical ranges in thermal analyses, eg. cold to room-temperature and room-temperature to hot ambient.

Density

Nominal density shall be a unique scalar value defined by M&P consistently with the nominal ply thickness.

DERIVATION OF DESIGN DATA FOR FIBROUS COMPOSITES

Maximum Operational Limit

The Maximum Operational Limit, or MOL, is defined as the maximum service temperature at which the material can operate continuously with neither suffering irreversible degradation, nor a massive loss of mechanical properties. Its value shall be defined by M&P.

3 PLY STIFFNESS

3.1 IN-PLANE ELASTIC MODULI

Shear modulus (G_{12})

No further considerations in addition to those presented in chapter 1480. The B-basis correction for this specific property shall not be required (see later discussion regarding B-basis correction for elastic moduli in buckling calculations).

Poisson's ratio (ν_{12})

The Poisson's ratio ν_{12} shall always be used as mean value (no need of B-basis design value for it).

Transverse elastic modulus in UD-Tape materials (E_2)

For the lamina/ply (uni-directional laminate), test reporting will show values of the elastic moduli in tension (E_{1t} and E_{2t}) and compression (E_{1c} and E_{2c}) tests. Typically, differences shall not be big, although not negligible (maybe up to a 15%).

For most of the structural analyses methodologies, composite laminates are modelled as linear elastic materials, i.e. with only one value for the elastic modulus no matter if material is in tension or compression. The starting point for setting the design values for E_1 and E_2 shall be done by calculating the average value between tension and compression:

$$E_1^0 = (E_{1t} + E_{1c}) / 2 \quad E_2^0 = (E_{2t} + E_{2c}) / 2$$

For UD-Tape materials, the final design value for the transverse modulus E_2 shall be finally that same average between tension and compression:

$$E_2 = E_2^0 = (E_{2t} + E_{2c}) / 2$$

As for G_{12} , the B-basis correction for E_2 in UD-Tape materials shall not be set from the tests data on the unidirectional laminate.

Longitudinal elastic modulus

The elastic modulus of any of the multi-axial laminates tested during the qualification tests program (average between values in tension and compression tests) may not fit perfectly the projection done using the elastic moduli of the ply and the Classical Lamination Theory (CLT):

$$E_x = (E_{xt} + E_{xc}) / 2 \neq E_x = f(E_1^0, E_2^0, G_{12}^0, \nu_{12})$$

DERIVATION OF DESIGN DATA FOR FIBROUS COMPOSITES

Typically the CLT method with the initial ply moduli from UD-laminates tests will overestimate the stiffness of laminates with low proportion of fibres aligned with load direction up to the quasi-isotropic (QI) laminate.

The policy proposed to derive the applicable design values for the elastic moduli shall be to use directly the values obtained in UD tests for the resin-driven parameters (i.e. set $G_{12} = G_{12}^0$, and for UD-tape materials, set $E_2 = E_2^0$), and then adjust the fibre-driven parameters (E_1 and, for bi-axial fabrics E_2 also) to minimize the error observed in predictions with tests data on uni-directional (only longitudinal/fibre direction) and multi-axial laminates, in tension and compression. Note that, for bi-axial fabrics (plain-weave, N-harness-satin, ...) the selected design value for E_1 and E_2 shall be one unique value ($E_1 = E_2$).

It is expected that the deviation between tests-based (average tension-compression) and calculated Young's moduli, using these corrected ply moduli, will not exceed a 5%. The calculated values would then properly approximate the laminates stiffness within the design space.

Note that, according the recommended policy introduced in chapter 1480 for normalizing across environments, the design values for elastic moduli shall be:

- ✓ resin-driven G_{12} (and E_2 for UD-Tape) at 70°C/Wet conditions,
- ✓ Poisson's ratio ν_{12} the average across data on all available environments,
- ✓ Young modulus in fibre direction E_1 (and also E_2 for woven fabrics) shall be a "fictitious" optimal value resulting after minimizing the error across environments (from RT/Dry to H/W) and across laminates (from those with lower fibre content in test load direction, through the QI, and up to the tested directed laminates and the UD in longitudinal direction)

Discussion above defines the standard process for obtaining the elastic moduli to use in structural analyses. However, in case other Airbus divisions have used previously the material in their airframes, Airbus DS may decide to use their design values for convenience.

Elastic moduli as B-basis for buckling stability analysis

The B-basis correction factor K_{BM} to be used to correct the nominal values of the elastic moduli (E_1 , E_2 and G_{12}) in buckling stability analysis shall be obtained from compression tests on multi-angular and UD (longitudinal) laminates, across all environmental conditions, dividing the B-basis measured modulus by the nominal value calculated with the Classical Lamination Theory. The minimum among these ratios shall be the final K_{BM} .

Estimations of resin-driven elastic moduli

In absence of test data for resin-driven elastic moduli, these could be estimated using Tsai & Melo "trace" approach (see Ref. 5) assuming the material belongs to the family of the carbon-fibre epoxy-resin-matrix systems:

	E_2	G_{12}	ν_{12}
UD-Tape	$0.0593 \cdot E_1$	$0.0356 \cdot E_1$	0.32
Bi-axial fabric	E_1	$0.0720 \cdot E_1$	0.05

DERIVATION OF DESIGN DATA FOR FIBROUS COMPOSITES

3.2 THROUGH-THICKNESS ELASTIC MODULI

Tests data for elastic moduli in through-thickness direction and planes shall not be normally available. Methods proposed in Ref. 3 and Ref. 4 can be used to estimate the values of E_3 , G_{13} , G_{23} , ν_{13} and ν_{23} .

- UD tapes (transversely isotropic)

$$E_3 = E_2$$

$$\nu_{13} = \nu_{12}$$

$$G_{23} = E_2 / [2 \cdot (1 + \nu_{23})]$$

ν_{23} as function of in-plane properties E_1 , E_2 , G_{12} , ν_{12} as proposed by Kuo (Ref. 3)

(*) Note that the Poisson ratio in an isotropic material (plane 2-3) is not expected to be above 0.5

- Woven fabrics

E_3 , G_{13} (= G_{23}) and ν_{13} (= ν_{23}) are estimated as proposed by Akkerman (Ref. 4)

Airbus DS in-house software *G23UD* and *frpGz* can be used for this purpose.

4 PLY STRENGTH

4.1 PLAIN STRENGTHS

Ultimate stresses

No further considerations in addition to those presented in chapter 1480.

Allowable strains

As discussed in chapter 3403 "PLAIN STRENGTH OF COMPOSITE LAMINATES", the allowable strains in tension and compression shall be obtained from the test data available from material qualification, for the unidirectional laminate and for multi-angular laminates. Typically at least the QI laminate should have been tested, but validation of the design allowables with test data also on multi-angular laminates is highly desirable, especially for UD-tape materials in tension.

4.2 INTERLAMINAR PLY STRENGTHS

Ultimate stresses

Values of F_{1t} , F_{1c} , F_{2t} , F_{2c} , and F_{12} shall be presented. No further considerations in addition to those presented in chapter 1480.

Fracture toughness

Values of G_{IC} and G_{IIC} shall be presented. No further considerations in addition to those presented in chapter 1480.

DERIVATION OF DESIGN DATA FOR FIBROUS COMPOSITES

4.3 OPEN HOLE AND BOLTED JOINTS

Values derived from the tests OHT, OHC, FHT, FHC and FBR (maximum before or at 6% deformation) in the QI laminate shall be presented. No further considerations in addition to those presented in chapter 1480.

Pull-out/through strength in a riveted laminate, with the fastener loaded in tension, can be normalized as the allowable through-thickness mean shear stress in the laminate, across a cylindrical section of the same diameter as the rivet-head or nut, caused by a tension force in the fastener:

$$F_{PO} = P_{PO} / (\pi \cdot D_h \cdot t)$$

...where:

- ✓ P_{PO} is the ultimate tension force for the pull-out/pull-through laminate failure
- ✓ D_h is the outer diameter of the fastener-head (or nut/collar)
- ✓ t is the thickness of the laminate (measured values in tests)

Previous tests experience (see Ref. 6) shows this magnitude, normalized in such a manner, is not significantly sensitive to the laminate directionality sequence and works fine for fasteners with countersunk heads of different kinds. However, tests results showed sensitivity to laminate thickness for configurations with t/d ratios below 1. A suggested approximation by Ref. 6, in absence of dedicated tests data, using the interlaminar material shear strength (ILSS) as reference, is as follows:

$$F_{PO} = F_{13} \cdot \psi(t/d) \quad \dots \text{with: } \psi = 1.0 \text{ if } t/d \geq 1, \text{ and } \psi = 1+0.6 \cdot (t/d-1) \text{ otherwise}$$

Dedicated tests are strongly recommended for the material and fastener systems in consideration. The same statistical procedures considered for other (resin driven) material strengths shall apply for pull-out/pull-through also.

4.4 COMPOSITE DAMAGE TOLERANCE

The design values applicable to strength analysis of laminates including damages due to low velocity impacts (BVID) shall be obtained according the approximation models proposed in chapter 1421 "MODEL FOR RESIDUAL STRENGTH AFTER IMPACT IN COMPOSITE LAMINATES".

DERIVATION OF DESIGN DATA FOR FIBROUS COMPOSITES

5 APPENDICES

STANDARD TEST METHODS

Typical tests methods to obtain the mechanical properties of the material referred through this chapter (most are standards in qualification programmes of fibrous composites within Airbus), are listed in following table:

Reference	Issue / Rev.	Title
ASTM E228	17	Standard Test Method for Linear Thermal Expansion of Solid Materials With a Push-Rod Dilatometer
ASTM E 831	14	Standard Test Method for Linear Thermal Expansion of Solid Materials by Thermomechanical Analysis
EN 2561	1995	Aerospace series. Carbon fibre reinforced plastics. Unidirectional laminates. Tensile test parallel to the fibre direction.
EN 2597	1998	Aerospace series. Carbon fibre reinforced plastics. Unidirectional laminates. Tensile test perpendicular to the fibre direction.
EN 2850	2018	Carbon fibre thermosetting resin. Unidirectional laminates. Compression test parallel to fibre direction.
AITM1-0002	3	Fibre Reinforced Plastics. Determination of in-plane shear properties ($\pm 45^\circ$ tensile test)
EN 2563	1997	Aerospace series. Carbon fibre reinforced plastics. Unidirectional laminates. Determination of the apparent interlaminar shear strength.
AITM1-0069	1	Fibre Reinforced Plastics. Determination of curved-beam failure load.
AITM1-0005	4	Carbon fibre reinforced plastics. Determination of interlaminar fracture toughness energy – Mode I (G _{IC} Test)
ASTM D5528	13	Standard Test Method for Mode I Interlaminar Fracture Toughness of Unidirectional Fiber-Reinforced Polymer Matrix Composites
AITM1-0006	2	Carbon fibre reinforced plastics. Determination of interlaminar fracture toughness energy – Mode II (G _{IIc} Test)
ASTM D6671	13	Standard Test Method for Mixed Mode I-Mode II Interlaminar Fracture Toughness of Unidirectional Fiber Reinforced Polymer Matrix Composites
AITM1-0007	5	Fibre Reinforced Plastics Determination of Plain, Open Hole and Filled Hole Tensile Strength
AITM1-0008	8	Fibre Reinforced Plastics. Determination of Plain, Open Hole and Filled Hole Compression Strength
AITM1-0009	3	Fibre Reinforced Plastics. Determination of Bearing Strength by either Pin or Bolt Bearing Configuration.
AITM1-0067	1	Fibre Reinforced Plastics Determination of Tension through the hole Strength on Fastened Joints

Table 1. *Mechanical tests methods (continues of following page...)*

DERIVATION OF DESIGN DATA FOR FIBROUS COMPOSITES

Reference	Issue / Rev.	Title
AITM1-0066	1	Fibre Reinforced Plastics. Determination of Pull-Through Strength of mechanical Joints.
ASTM D7332	16	Standard Test Method for Measuring the Fastener Pull-Through Resistance of a Fiber-Reinforced Polymer Matrix Composite
AITM1-0010	3	Fibre Reinforced Plastics. Determination of Compression Strength After Impact.
ASTM D7137	12	Standard Test Method for Compressive Residual Strength Properties of Damaged Polymer Matrix Composite Plates
TEA-T-SP-170011	Is. A	TEST METHOD FOR DETERMINATION OF THE STRENGTH AFTER IMPACT OF SOLID LAMINATES MADE OF FIBRE REINFORCED PLASTICS

Table 1. *Mechanical tests methods (... from previous page)*

ABBREVIATIONS AND SYMBOLS

Symbol	Units	Definition
BVID	-	Barely visual impact damage
CAI	MPa / $\mu\epsilon$	Compression After Impact strength of laminate including an undetected damage
CLT, CLA	-	Classical Laminate Theory/Analysis
CPT	mm	Cured Ply Thickness
D _h	mm	Rivet head (or nut) external diameter
E	mm	Elastic (Young's) modulus
E-KDF, K _{E-KDF}	-	Environmental Knock Down Factor: ratio between the value of a mechanical property obtained at H/W conditions and the corresponding value at RT/AR conditions
DV	?	Design Value: value of a mechanical property to be used in structural analyses
E ₁	MPa	Elastic modulus in 0°/warp material direction
E _{1c}	MPa	Compression elastic modulus in 0°/warp material direction
E _{1t}	MPa	Tensile elastic modulus in 0°/warp material direction
E ₂	MPa	Elastic modulus in 90°/weft material direction
E _{2c}	MPa	Compression elastic modulus in 90°/weft material direction
E _{2t}	MPa	Tensile elastic modulus in 90°/weft material direction
E _x	MPa	Elastic modulus (theoretical) of laminate in load test direction
F ₁₂	MPa	Ultimate in-plane shear stress
F ₁₃	MPa	Ultimate interlaminar shear stress (ILSS)
F _{1c}	MPa	Ultimate compression stress in 0°/warp material direction

DERIVATION OF DESIGN DATA FOR FIBROUS COMPOSITES

F_{1t}	MPa	Ultimate tensile stress in 0°/warp material direction
F_{2c}	MPa	Ultimate compression stress in 90°/weft material direction
F_{2t}	MPa	Ultimate tensile stress in 90°/weft material direction
F_{PO}	MPa	Ultimate Pull-Out through-thickness shear stress
FBR	MPa	Bearing strength (at 6% deformation)
FHC	MPa	Filed Hole Compression strength
FHT	MPa	Filed Hole Tension strength
P_{PO}	N	Pull-Out tension strength
FRP	-	Fibre Reinforced Polymer
G_{12}	MPa	In-plane shear modulus
G_{IC}	N/mm	Fracture toughness in mode I (opening)
G_{IIC}	N/mm	Fracture toughness in mode II (shear)
ILSS	MPa	Interlaminar Shear Strength
K_B	-	Ratio between the B-basis value and the mean-value of a data sample or population
OHC	MPa	Open Hole Compression strength
OHT	MPa	Open Hole Tension strength
PC	MPa	Plane Compression modulus/strength
PT	MPa	Plane Tension modulus/strength
QI	-	Quasi-Isotropic, laminate with in-plane behaviour similar to isotropic materials, typically built with the same number of plies at 0°, 90°, 45° and -45° directions, stacking sequences of type ((45/-45/90/0) n)S, with n a positive integer number.
SW	-	Software
t	mm	Thickness (time in sub-chapter about dimensional analysis)
T	°C	Temperature
TAI	MPa / $\mu\epsilon$	Tension After Impact strength of laminate including an undetected damage
TBD	-	To Be Defined
UD-tape	-	Uni-directional tape FRP material
ϵ_{1c}	-	Ultimate compression strain in 0°/warp material direction
ϵ_{1t}	-	Ultimate tensile strain in 0°/warp material direction
ϵ_{2c}	-	Ultimate compression strain in 90°/weft material direction
ϵ_{2t}	-	Ultimate tensile strain in 90°/weft material direction
ν_{12}	-	Poisson' s ratio
γ_{12}	-	Ultimate in-plane shear strain

DERIVATION OF DESIGN DATA FOR FIBROUS COMPOSITES

REFERENCES

	<i>Document Reference</i>	<i>Issue</i>	<i>Title</i>
Ref. 1	[FAA/SAE] CMH-17	03/12	COMPOSITE MATERIALS HANDBOOK
Ref. 2	TAE-T-MM-170009	Is. C	GUIDELINES FOR RESULTS ANALYSIS OF COUPONS TESTS IN COMPOSITE MATERIALS
Ref. 3	[Y. M. Kuo, H. J. Lin et al]	2007	Estimating the Elastic Modulus through the Thickness Direction of a Uni-direction Lamina which Possesses Transverse Isotropic Property
Ref. 4	[Remko Akkerman]	2004	Laminate mechanics for balanced woven fabrics
Ref. 5	[S.W.Tsai and J.D.D.Melo]	2015	Composite Materials Design and Testing. Unlocking mystery with invariants.
Ref. 6	NT-L-ADP-04002	Is. D	Allowable pull-out load. Summary of test results for Z-19.775/Z-19.780 (AS4/8552)

SOFTWARE

<i>Program</i>	<i>Version</i>	<i>Description</i>
COMALL	0.13	COMposite ALLOWables software calculates the design values for a fibrous composite material, processing the results of a standard coupons tests program
G23UD	1.2	Estimation of G23 shear modulus in fibrous composites with transverse isotropy (uni-directional)
frpGz	1.1	Estimation of thorough thickness shear modulus in fibrous composites

Table 2. *Software programs referenced in this chapter*

RECORD OF REVISIONS

<i>Revision Date</i>	<i>Authors</i>	<i>Change Reason</i>	<i>Pages/Sub-chapters affected</i>
1.0 06/08/2020	G. Baños	First issue	All pages
1.1 15/02/2023	G. Baños	Added warning note about transverse Poisson ν_{23} not expected to be above 0.5	Page 4
1.2 08/06/2023	G. Baños	Clarify convention to derive the bearing allowable strength of bolted joints	Page 5

Table 3. *Record of Revisions: authors, change reasons and sections/pages affected*

DERIVATION OF DESIGN DATA FOR FIBROUS COMPOSITES

APPROVAL SIGNATURES TABLE

<i>Dpt. \ Site</i>	<i>Getafe</i>	<i>Manching</i>
Stress Methods (TEPAP-TL4)	G. Baños 08/06/2023	-
Stress Analysis (TEPAA)	-	F. Glock 29/06/2023

Table 4. *Approval signatures table for last revision of this chapter*